E-RAKSHAK: SURAT SMART TRAFFIC COMMAND

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5-Minute SUMO Micro-Simulation & Adaptive Traffic Analytics Report

Corridor: Surat Arterial Spine (SVNIT -> Ghod Dod -> Majura Gate -> Sahara Darwaja) | **Policy:** Adaptive Traffic Control (ADAPTIVE) | **Demand:** PEAK (90.0 veh/min) | **Duration:** 300s

3085.0 vph	**28.2** km/h	**1.6** sec	**257** trips
THROUGHPUT CAPACITY	CORRIDOR SPEED	INTERSECTION DELAY	COMPLETED TRIPS

1. What-If Scenario Comparison (Adaptive Control vs Fixed-Time Baseline)

Throughput Rate	2344.6 veh/hr	**3085.0 veh/hr**	**+31.6%**
Average Corridor Speed	20.3 km/h	**28.2 km/h**	**+38.9%**
Average Intersection Delay	2.5 s	**1.6 s**	**-36.0%**
Completed Corridor Trips	194 trips	**257 trips**	**+31.6%**

2. Corridor Bottleneck Hotspots Ranking

Rank	Intersection Location	Score	Avg Delay	Severity	Primary Contributing Factor
	Majura Gate BRTS Multi-Leg Hub		52.8s	**Critical**	High cross-turning volume & BRTS lane friction
#2	**Sahara Darwaja Railway Flyover**	100/100	64.1s	**Critical**	High cross-turning volume & BRTS lane friction
	Ghod Dod Road Commercial Cross		38.2s	**Critical**	Commercial curb friction
#4	**SVNIT / Ichchhanath Circle**	50/100	24.5s	**High**	Commercial curb friction

3. Actionable Engineering Recommendations

Code	Category	Recommendation & Action Plan	Impact
	Signal Timing	**Optimize Arterial Progression Offset (Green Wave)** Progressive green signal offset along J_SVNIT -> J_MAJURA at 45 km/h progression speed to reduce corridor stop delay by ~32%.	
REC-02	Corridor Infrastructure	**Install Continuous Median Delineators on BRTS Lane L0** Detected 256 illegal mixed vehicle intrusions along BRTS lane L0. Physical flexible bollards recommended to protect rapid transit speed.	Critical

	Lane Management	**Designate Dedicated Right-Turn Slip Lane on Lane 2** Right-turning traffic blocks through-movements during peak hours. Dedicated slip lane will decouple straight-line throughput.	